

# **VANTRA**

(Vehicle ANPR Trajectory)

Project Proposal for UCS503P

**Submitted to:**

Ms. Nisha Thakur

Computer Science and Engineering Department  
Thapar Institute of Engineering and Technology, Patiala

**Submitted by:**

1024030936 Daksh Jand

1024030332 Saksham

1024030837 Raunak Mahajan

**Group:** 8, BE Third Year, COE

# 1. Introduction

## 1.1 Background and Context

Automatic Number Plate Recognition (ANPR) systems can identify vehicles from camera images. However, in a large road network, a vehicle may pass through multiple cameras, with each camera independently storing its observation. This makes it difficult to reconstruct the complete route followed by a vehicle.

**VANTRA (Vehicle ANPR Trajectory)** is proposed to address this gap by connecting individual camera sightings into an ordered vehicle trajectory. Instead of depending only on the license plate number, VANTRA combines multiple signals such as plate text, OCR confidence, vehicle appearance, camera connectivity, route plausibility and travel-time information.

The central principle of VANTRA is **explainability**. Every connection between two vehicle sightings should be supported by evidence that can be inspected. The system is designed as a modular Software Engineering solution, allowing components such as OCR and vehicle re-identification to be improved or replaced independently.

## 1.2 Problem Statement

Existing camera networks can produce separate records when the same vehicle passes through multiple cameras. OCR errors, missed detections and visually similar vehicles can make it difficult to determine whether two sightings belong to the same vehicle.

For example, a vehicle travelling through Camera A, Camera B and Camera C may generate three disconnected records rather than one continuous journey.

VANTRA aims to provide an integration layer that evaluates multiple pieces of evidence before connecting two sightings. When the evidence is insufficient, the system can avoid forcing a connection. Similarly, if a vehicle would need to travel between two cameras in an impossible amount of time, the transition can be flagged as an anomaly.

# 2. Project Objectives

- Create a common pipeline for camera detections, plate information, vehicle appearance and trajectory reconstruction.
- Use OCR to extract license-plate text while retaining OCR confidence.
- Handle noisy OCR readings using exact and fuzzy matching.
- Use route plausibility and travel-time constraints to prevent unrealistic trajectory links.
- Use vehicle appearance as supporting evidence when plate information is weak or ambiguous.
- Provide a GIS dashboard for plate search, trajectory visualization, alerts and evidence review.
- Provide explainable evidence for each trajectory connection.
- Evaluate the system using automated tests, stress tests, scale testing and held-out OCR experiments.

# 3. Proposed Solution

## 3.1 Overview

VANTRA will implement a pipeline-based system in which camera observations are converted into useful signals and then evaluated by an explicit trajectory decision layer.

- Camera and Detection Pipeline
- OCR and Plate Processing
- Vehicle Appearance Analysis
- Trajectory Matching Engine
- Database and Spatial Data Layer

- Alert Engine
- GIS Dashboard

### 3.2 Core Workflow

- 1. Camera observations enter the system.
- 2. License plates are localized and OCR is used to extract plate text.
- 3. OCR confidence and vehicle appearance information are retained.
- 4. The system compares the current observation with possible previous sightings.
- 5. Exact plate matching is attempted first.
- 6. If necessary, fuzzy matching is used to handle OCR differences.
- 7. Vehicle appearance is used to resolve ambiguous candidates.
- 8. Route connectivity and travel-time constraints are checked.
- 9. A valid connection is added to the vehicle trajectory.
- 10. Physically impossible transitions are flagged as anomalies.
- 11. The resulting trajectory and evidence are stored and displayed through the GIS dashboard.

## 4. System Architecture

VANTRA follows a modular pipeline architecture. The camera pipeline processes observations and generates plate text, OCR confidence, vehicle appearance, vehicle type/colour, camera and timestamp information. The decision layer then combines these signals with route and travel-time constraints.

The prototype evaluation uses an 18-camera simulated Bengaluru network with approximately 23,000 historical detections over 21 days. The simulator introduces OCR errors, missed cameras, dwell periods and congestion to test the trajectory system under imperfect observations.

### 4.1 Trajectory Matching

| Tier | Operation                                |
|------|------------------------------------------|
| 1    | Exact plate + route plausibility         |
| 2    | Fuzzy unique plate matching              |
| 3    | Fuzzy plate + vehicle appearance         |
| 4    | Vehicle appearance + route plausibility  |
| 5    | Impossible travel-time anomaly detection |

## 5. Technology Stack

| Component       | Technology                         |
|-----------------|------------------------------------|
| Backend         | Python, FastAPI                    |
| Frontend        | React, Vite                        |
| Map/GIS         | Leaflet                            |
| Database        | PostgreSQL, PostGIS                |
| OCR             | EasyOCR                            |
| Plate Detection | YOLOv8                             |
| Data Analysis   | NumPy, Pandas, Matplotlib, Seaborn |
| Deployment      | Docker, Docker Compose             |

|         |                               |
|---------|-------------------------------|
| Testing | Pytest, Integration Tests, CI |
|---------|-------------------------------|

## 6. Machine Learning Components

### 6.1 EasyOCR

EasyOCR converts cropped license-plate images into text. The project uses a CRAFT + CRNN pipeline. The OCR recognizer was fine-tuned using real license-plate data containing 7,318 real plates, with augmentation producing 29,272 training rows.

The OCR training is supervised learning because the training images are paired with correct plate-text labels. The fine-tuning process also represents transfer learning because the starting recognizer was already pretrained.

### 6.2 YOLOv8

YOLOv8 is intended to detect and localize the license plate within a vehicle image by generating a bounding box. It does not perform character recognition; OCR performs that task. The current prototype uses a contrast-based localization stand-in, while a fine-tuned YOLOv8 model trained on Indian plate footage is planned as a production upgrade.

## 7. Dashboard and Features

- Plate Search
- Trajectory View
- Heatmap
- Live Replay
- Alerts
- Compare
- Audit
- Tuning
- Case File Export

## 8. Database and Data Management

PostgreSQL will store structured system information, while PostGIS will provide spatial database capabilities. The database will contain camera locations, camera connections, vehicle detections, plate information, travel-time statistics, reconstructed trajectories, alerts and evidence associated with trajectory hops.

For larger-scale deployment, the architecture can be extended using Kafka for ingestion, GPU inference workers, TimescaleDB or ClickHouse for high-volume detections, object storage for image crops and vector indexing for appearance search.

## 9. Evaluation and Testing

Testing will be performed at multiple levels: unit testing for trajectory matching and alert behaviour; integration testing for API and database services; OCR stress testing under blur, angle, low-light, occlusion and rain; trajectory testing with missed cameras; anomaly testing with impossible vehicle transitions; and scale testing using a larger detection database.

The project documentation reports 71 automated tests with 88% coverage.

### 9.1 Evaluation Results

- Clean synthetic OCR: 98% exact
- Blur stress OCR: 42% exact

- Angle stress OCR: 52% exact
- Low-light OCR: 95% exact
- Occlusion OCR: 83% exact
- Rain OCR: 95% exact
- Zero missed cameras: 100% full-path reconstruction
- One missed camera: 100% full-path reconstruction
- Two missed cameras: 80% full-path reconstruction
- Injected cloned traversals: 15/15 flagged

These synthetic results validate the prototype logic and should not be interpreted as city-deployment accuracy.

## 10. Real-World OCR Evaluation

A separate held-out experiment used 1,047 real photographs from US/EU and Taiwan plate datasets. Fine-tuned EasyOCR achieved 47.9% exact plate accuracy and 78.3% character accuracy. The updated Indian-domain evaluation reported approximately 50.3% exact accuracy and 87.1% character accuracy on 312 held-out Indian images from 152 unique plates.

These results demonstrate improvement from fine-tuning but also identify real-world OCR accuracy as an important limitation of the current system.

## 11. Scalability

A separate scale test used 120 cameras and 1.2 million detections. The documented median trajectory reconstruction latency for the busiest plate was approximately 1.5 seconds, with wildcard plate search at approximately 2.6 ms and heatmap queries at approximately 38 ms.

For future city-scale deployment, the system can move towards Kafka-based asynchronous ingestion, GPU inference workers, real road-network data, high-volume databases and distributed processing.

## 12. Project Scope and Deliverables

### 12.1 Initial Deliverables

- Camera observation pipeline
- Plate localization and OCR processing
- Trajectory matching engine
- PostgreSQL/PostGIS database
- REST APIs using FastAPI
- GIS dashboard
- Alert engine
- Automated testing
- Docker-based deployment

### 12.2 Subsequent Deliverables

- Improved Indian license-plate detection
- Expanded OCR training dataset
- Perspective correction for oblique plates
- Super-resolution for small plate crops

- Improved vehicle Re-ID model
- Real road-network integration
- Asynchronous streaming architecture
- Enhanced authentication, authorization, audit and privacy controls

## 13. Risks and Mitigation

| Risk                       | Mitigation                                                   |
|----------------------------|--------------------------------------------------------------|
| OCR errors                 | Confidence-aware fuzzy matching and appearance evidence      |
| Missed camera detection    | Trajectory recovery across camera gaps                       |
| Visually similar vehicles  | Combine appearance with plate and route evidence             |
| False trajectory links     | Route and minimum travel-time constraints                    |
| Poor image quality         | Preprocessing, perspective correction and improved detection |
| Scalability issues         | Indexed databases and asynchronous processing                |
| Privacy concerns           | Authentication, authorization, audit and retention controls  |
| Synthetic-data limitations | Expand evaluation using diverse real-world Indian data       |

## 14. Future Work

- 1. Fine-tune YOLOv8 using a large and diverse Indian license-plate dataset.
- 2. Expand the Indian OCR dataset with different camera angles, distances and lighting.
- 3. Add super-resolution and image tiling for small plate crops.
- 4. Add perspective correction for oblique plates.
- 5. Replace the handcrafted appearance descriptor with a trained vehicle Re-ID model.
- 6. Use real camera GPS locations and a routable road graph.
- 7. Introduce Kafka-based ingestion and asynchronous GPU inference.
- 8. Strengthen security, privacy, authentication, authorization and data-retention mechanisms.

## 15. Conclusion

VANTRA proposes a modular Software Engineering solution for connecting isolated ANPR observations into an explainable vehicle trajectory. By combining OCR, vehicle appearance, route connectivity and travel-time constraints with explicit decision rules, the system is designed to handle imperfect camera observations without relying on a single signal.

The current prototype demonstrates multi-camera trajectory reconstruction on a controlled simulated network, while the evaluation identifies OCR accuracy and real-world deployment as important areas for future improvement. The project provides a practical engineering foundation that can be progressively extended towards larger and more realistic deployments.